

Luke Pitstick

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EDUCATION

University of Colorado Boulder
B.A. Computer Science, B.A. Political Science

Aug. 2024 – May 2028
Boulder, CO

RELEVANT EXPERIENCE

Junior Data Manager
National Oceanic and Atmospheric Administration / CIRES

Oct. 2025 – Present
Boulder, CO

- Build and maintain automated pipelines that ingest and validate **10+ TB/week** of ocean acoustic data into Oracle DB and AWS S3.
- Work with NOAA research vessel data providers to help them submit datasets, resolve metadata and QC issues, and put water column data into NCEI's archive.
- Build an internal tool that consolidates ingestion, QC, metadata validation, and archival into one workflow.

AI/ML Engineer Intern
Wattbyte Nexus

Jun. 2026 – Aug. 2026
Golden, CO

- Implemented power line galloping surrogate models, achieving an **RMSE of 3.1 cm and a speedup of five orders of magnitude** compared to classical finite element methods.
- Built event-driven wildfire-risk simulations using NOAA weather data, AWS, and Redis to model grid assets, environmental conditions, and failure scenarios.
- Wrote customer docs on swapping NWS weather data for their own feeds.

Data Science Intern
Ranked Choice Voting for Longmont

Aug. 2025 – Dec. 2025
Remote

- Created **demographic maps** with ArcGIS, R, and Python, enabling campaign teams to identify turnout gaps and execute targeted outreach.
- Developed fixed-effects regression models to predict voter turnout in Longmont, to prioritize precincts for outreach efforts.

LEADERSHIP EXPERIENCE

Representative-at-Large & Press Secretary
University of Colorado Student Government

May 2025 – May 2026
Boulder, CO

- Reviewed budget proposals for a **\$36M** student budget across 12 cost centers, coordinating with university administrators to ensure equitable fund distribution.
- Led a 6-person team to increase follower counts across social media channels by **15%**.

PROJECTS

Road Crack Detection Engine | *Python, LightGlue, PyTorch, Modal, React* | [GitHub](#)

Apr. 2026 – Present

- Built a road survey platform using React, Modal, and Python that runs computer vision models on survey footage to return localized damage classifications and severity estimates.
- Trained a custom AI pipeline on a 50,000+ image dataset to detect, classify, and calculate the severity of road damage using YOLO26-seg, resulting in **80% precision and 75% mAP@50**.
- Developed a temporal engine to measure crack growth over time using a feature matching algorithm built with LightGlue.

Renewably | *Python, XGBoost, Polars, React, FastAPI* | [Demo](#)

Apr. 2026

- Won 1st place at BlasterHacks 2026; built an interactive renewable energy siting platform with React and ArcGIS.
- Trained an XGBoost model on geospatial features to score wind-farm site viability against existing wind turbine locations. Achieved **96% accuracy** on validation data.
- Developed a FastAPI-powered optimization pipeline capable of evaluating thousands of candidate locations within user-defined regions, achieving **inference times under 1 minute**.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, SQL, R

Frameworks/Libraries: PyTorch, NumPy, scikit-learn, pandas, Polars, ggplot2, Matplotlib, React, FastAPI

Cloud/Databases: AWS, GCP, Oracle DB, PostgreSQL

Other: Agile, Figma, ArcGIS, Bash, Linux, Git